

ACT: Anchor-Calibrated Trust Estimation When Corrupted Sources Form the Majority

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Abstract—Pairwise agreement is often used to estimate source reliability. It may not work as intended when the majority of the sources are corrupted. The system may trust the untrusted cohort because a sizable corrupted cohort may agree more with one another than with trusted sources. We present Anchor-Calibrated Trust (ACT), a semi-supervised trust-orientation estimator. ACT estimates which cohort we should trust based on pairwise agreement and a small labeled anchor set. The anchor is used only for orientation, not to train the underlying reward models. We evaluated ACT on two datasets, namely SST-2 sentiment classification with BERT models and HH-RLHF preference modeling with GPT-2-XL+LoRA reward models. Under feedback with a trusted majority, ACT preserves the proper trust direction; under feedback with a corrupted majority, it corrects the trust orientation across both reported seeds. ACT-based orientation fixes the same flaw that shows up in five other reliability techniques, such as GLAD and Dawid-Skene. ACT maintains the correct corrupted-majority orientation in at least 97.5% of configurations under moderate random anchor-label noise.

Index Terms—Trust estimation, corrupted-majority feedback, anchor-based calibration, semi-supervised learning, reward model aggregation, RLHF

I. INTRODUCTION

Modern learning systems mix signals from flawed sources, including human annotators, reward models, and AI judges. When such feedback are aligned, averaging or majority voting can turn coordinated errors into confident downstream decisions. This risk is acute in high-stakes settings such as power grids, where AI models may inform forecasting, anomaly detection, and control while sharing training data or assumptions. Their errors are correlated rather than independent and a wrong aggregate can propagate into physical consequences.

Pairwise agreement is often used as an unsupervised proxy for source reliability, assigning greater trust to sources that agree more frequently with the pool. Related unsupervised ensemble methods estimate or rank source accuracy from patterns of agreement and covariance without labeled data [1], [2]. Such methods rely on assumptions about source independence; strong dependencies among predictors can substantially affect reliability estimation [3]. Under *corrupted-majority feedback*, however, a larger, coherent corrupted cohort may receive higher agreement than the trusted cohort.

We introduce *Anchor-Calibrated Trust* (ACT), a semi-supervised trust-orientation estimator. ACT initially uses pairwise agreement to estimate a two-cohort source structure. It then evaluates the two cohorts on a small labeled anchor set and treats the more accurate cohort as trusted. The anchor is used explicitly to orient the agreement-derived partition; it does not train the underlying classification or reward models.

ACT operates upstream of aggregation by converting source scores into an oriented reliability map that downstream methods can use as source-level reliability weights, provided that agreement and anchor accuracy are well defined and the two-cohort assumptions hold. We evaluate ACT with three reliability-aware aggregators from COBRA [4], which serves only as the downstream evaluation context.

ACT is assessed in two controlled multi-source settings: SST-2 sentiment classification with BERT models and HH-RLHF preference modeling with GPT-2-XL+LoRA reward models. Across two source-construction seeds, Agreement is inverted in every main corrupted-majority configuration, whereas ACT recovers the Declared-trust binary partition and preserves the existing orientation in every trusted-majority configuration. The corrected reliability maps produce substantial downstream gains across all three evaluated aggregators. We also test whether this orientation failure is limited to pairwise Agreement. Dawid-Skene, GLAD, spectral estimation, truth discovery, and a Snorkel-family label model show the same corrupted-majority inversion in our experiments, while anchor-based orientation restores the correct direction. The evaluation measures recovery of the binary trusted/corrupted partition, not continuous calibration or within-cohort ranking. Our contributions include:

- 1) **Characterizing trust-orientation** inversion under corrupted-majority feedback: We show that, when corrupted sources form an internally coherent majority, pairwise agreement can recover the two-cohort structure while assigning higher reliability to the corrupted cohort. We reproduce the same orientation failure across five other reliability-estimation methods.
- 2) **Anchor-Calibrated Trust**: We introduce ACT, which

partitions sources by agreement and then uses a small labeled anchor set to orient the partition, without training the underlying models.

- 3) **Empirical evaluation:** We evaluate ACT on two substrates with two seeds each, across anchor budgets, random anchor-label noise, and source-count sub-sampling, and add stress tests covering threshold sensitivity, the previously excluded parity region, graded and multi-cohort source quality, correlated corruption, and targeted or worst-case anchor perturbations.

II. RELATED WORK

Source reliability and label inference: Multi-annotator methods jointly infer latent labels and source reliability. Dawid-Skene estimates annotator-specific error rates through a latent-class model [5], while GLAD models annotator expertise and item difficulty [6]. Raykar et al. jointly learn a predictive model while estimating annotator reliability [7], and Karger et al. develop iterative inference methods for repeated crowd sourced labels [8]. Truth-discovery methods similarly estimate source quality and the truth of conflicting claims [9], while weak-supervision systems such as Snorkel learn from noisy labeling functions by modeling their accuracies and dependencies [10]. ACT addresses a narrower orientation problem because it neither infers labels for the full dataset nor estimates a complete source model. Given a sufficiently separated agreement-derived two-cohort partition, it uses labeled anchors only to determine which cohort should be trusted.

Learning with trusted examples: Clean examples have been used to estimate label-corruption processes or correct training objectives under label noise [11]. ACT uses a labeled set, but the anchors neither retrain the source models nor estimate an item-level noise transition matrix. They serve only to orient the agreement-derived source partition. We also test whether directly providing these labels to latent-source estimators is enough to resolve the corrupted-majority orientation problem.

Robust and Byzantine aggregation: Byzantine-robust learning develops aggregation rules for settings in which distributed workers may return faulty or adversarial updates [12], [13]. ACT instead operates on decisions from trained sources and outputs a reliability map for a score aggregator. It provides no Byzantine-robust optimization guarantee, and our targeted and worst-case anchor tests should be viewed as stress tests rather than an adversarial-robustness guarantee.

Reward modeling and downstream aggregation: Preference-based reward models use human comparisons to learn reward signals for reinforcement learning from human feedback [14]–[16]. We use multiple reward models as one evaluation substrate; ACT does not modify reward-model training or policy optimization. COBRA introduced the Root-of-Trust (RoT), Dynamic Reliability Weighted Aggregation (DRWA), and Adaptive Variance-Guided Attention (AVGA) aggregation strategies [4]. ACT addresses the upstream problem of estimating trust orientation, while these aggregators provide the downstream evaluation context.

III. PROBLEM FORMULATION

We separate source trust estimation from downstream aggregation. A trust estimator determines the influence assigned to each source, while an aggregator combines the resulting weighted source outputs. This separation allows failures in trust orientation to be distinguished from failures of the aggregation rule. Consider N sources evaluated on a common set of items. For item t , source i produces a score vector $\mathbf{s}_{i,t}$. A trust estimator maps observed source behavior to a reliability vector $\mathbf{r} = (r_1, \dots, r_N)$, where r_i controls the downstream influence of source i . A downstream aggregator G , such as RoT, DRWA, or AVGA, then produces $\hat{y}_t = G(\mathbf{s}_{1,t}, \dots, \mathbf{s}_{N,t}; \mathbf{r})$. The interpretation of $\mathbf{s}_{i,t}$ and the final decision rule depend on the substrate and are defined in Section V. Reliability-aware aggregation is useful only when $\mathbf{r}$ is correctly oriented. If trusted sources receive larger values, corrupted sources can be downweighted; if the map is inverted, the same aggregator amplifies them. We compare four trust conditions:

- **Equal trust:** All sources receive the same reliability value, so the map contains no source-level trust information. Simple averaging is treated separately as the common downstream accuracy reference.
- **Declared trust:** The known trusted/corrupted assignments define a two-level oracle reliability map. This condition is used only as an evaluation reference, not available to ACT.
- **Agreement:** Each source is weighted by its mean pairwise decision agreement with the other sources. This map can recover a two-cohort structure without identifying which cohort is trusted.
- **Anchor-Calibrated Trust:** It uses Agreement to estimate the two-cohort structure and a small labeled anchor set to orient the resulting partition. It outputs a trusted/corrupted reliability map for downstream aggregation.

We evaluate these trust conditions with the RoT, DRWA, and AVGA aggregators from COBRA [4]. COBRA provides the downstream evaluation context and is not part of ACT. We separately evaluate whether the same corrupted-majority orientation problem occurs with independent reliability estimators, including Dawid-Skene, GLAD, spectral estimation, truth discovery, and a Snorkel-family label model.

IV. ANCHOR-CALIBRATED TRUST

Anchor-Calibrated Trust (ACT) separates *structure discovery*, group sources with similar agreement patterns, from *trust orientation*, which group receives higher reliability. ACT uses pairwise agreement to identify two source cohorts and a small labeled anchor set to determine which cohort should be trusted. This distinction is necessary because, under corrupted-majority feedback, agreement may recover the cohort structure while assigning greater reliability to the corrupted cohort.

A. Agreement scores

Let $N \geq 2$ sources produce binary decisions on the same T items, with $d_i(t) \in \{0, 1\}$ denoting the decision of source

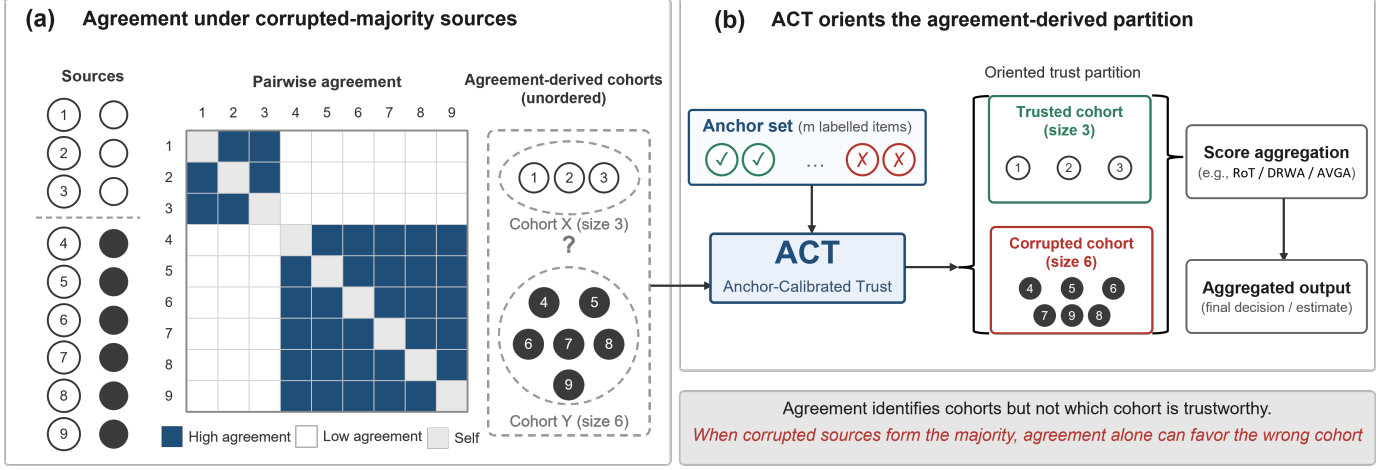


Fig. 1. (a) Pairwise agreement identifies two source cohorts but cannot determine which is trusted; under corrupted-majority feedback, it favours the corrupted cohort. (b) ACT uses labelled anchors to orient the agreement-derived partition before RoT, DRWA, or AVGA aggregation.

i on item t . Substrate-specific decision rules are defined in Section V. The agreement score of source i is

$$a_i = \frac{1}{(N-1)T} \sum_{j \neq i} \sum_{t=1}^T \mathbb{I}[d_i(t) = d_j(t)], \mathbf{a} = (a_1, \dots, a_N)$$

B. Structure discovery

Let π be the stable permutation satisfying $a_{\pi(1)} \leq \dots \leq a_{\pi(N)}$. For each $k \in \{1, \dots, N-1\}$, let $\bar{a}_{\text{low}}(k)$ and $\bar{a}_{\text{high}}(k)$ be the means of the bottom k and top $N-k$ agreement scores. ACT selects

$$k^* = \arg \max_{k \in \{1, \dots, N-1\}} k(N-k) (\bar{a}_{\text{high}}(k) - \bar{a}_{\text{low}}(k))^2$$

The resulting source sets are $C_{\text{low}} = \{\pi(1), \dots, \pi(k^*)\}$, $C_{\text{high}} = \{\pi(k^* + 1), \dots, \pi(N)\}$. These labels refer only to agreement magnitude; C_{high} is not assumed to be trusted. Membership is determined by position in the stable ordering, so equal agreement scores may fall on opposite sides of the split. Ties in the split objective are resolved in favor of the smallest k .

Agreement separation is $\Delta_s = \bar{a}_{\text{high}}(k^*) - \bar{a}_{\text{low}}(k^*)$

C. Anchor-based orientation

Anchor items are sampled from a labeled pool disjoint from the evaluation items. They are used only to orient the agreement-derived partition, not to train the underlying models. For m anchor items with labels y_ℓ , the anchor accuracy of source i is

$$q_i = \frac{1}{m} \sum_{\ell=1}^m \mathbb{I}[d_i^{\text{anc}}(\ell) = y_\ell]$$

For cohort C , define

$$\bar{q}(C) = \frac{1}{|C|} \sum_{i \in C} q_i, \quad \Delta_q = \bar{q}(C_{\text{high}}) - \bar{q}(C_{\text{low}})$$

Outside the threshold-sensitivity ablation, we use $\tau_s = \tau_a = 0.05$ as fixed fail-safe margins rather than tuned optima. Section VI evaluates sensitivity to both thresholds. ACT assigns

$$r_i = \begin{cases} 1 - a_i, & \Delta_s \geq \tau_s \text{ and } \Delta_q \leq -\tau_a, \\ a_i, & \text{otherwise} \end{cases}$$

Thus, ACT reverses Agreement only when the two cohorts are sufficiently separated and the anchors provide sufficient evidence that C_{low} is more accurate. If either check is inconclusive, or if the anchors support C_{high} , ACT retains the original Agreement map. This fallback avoids unsupported reversals but does not guarantee correct orientation under corrupted-majority feedback.

V. EXPERIMENTAL SETUP

We evaluate whether ACT recovers the declared trusted/corrupted partition and whether the resulting reliability map improves downstream aggregation. Source status is known because corruption is controlled, but these assignments are used only as an oracle evaluation reference and are not provided to ACT.

A. Evaluation Substrates

We use two nine-source substrates and evaluate both 3/9-corrupted (trusted-majority) and 6/9-corrupted (corrupted-majority) compositions.

SST-2 classification: The sentiment-analysis (SA) substrate contains nine independently trained BERT classifiers. For corrupted sources, 80% of training labels are inverted. Each composition is evaluated under two seeds. A source predicts the class with greater probability, and downstream accuracy is measured against the original SST-2 labels.

HH-RLHF preference modelling: The preference-learning (TRL) substrate contains nine independently trained GPT-2-XL reward models with LoRA adapters. For corrupted sources, the chosen and rejected responses are exchanged in 80% of

assigned training comparisons. Each composition is evaluated under two data-shuffle seeds. A source selects the response with greater reward under a common score convention, and downstream accuracy is the fraction of pairs for which the aggregate ranks the chosen response above the rejected response.

B. Trust Conditions and Aggregation

We compare four conditions: Equal trust, Declared trust, Agreement, and ACT. Equal trust assigns a common value to all sources. Declared trust is the oracle binary map. Agreement uses mean pairwise agreement. ACT orients the agreement-derived partition using labeled anchors.

For the anchor-budget ablation, we additionally evaluate a per-source anchor-only probe with $r_i = q_i$, where q_i is source i 's anchor accuracy. The probe uses the same anchor draws as ACT and is reported only in that ablation. Simple averaging ignores the reliability map and serves as the common no-trust baseline. RoT, DRWA, and AVGA from COBRA [4] consume the reliability map produced by the selected estimator.

To test whether corrupted-majority inversion is specific to Agreement, we also evaluate Dawid-Skene, GLAD, spectral estimation, truth discovery, and a Snorkel-family label model (triplet closed form) using the same binary source decisions. Their reliability outputs are evaluated directly using the trust-orientation metric in Section V-F, without requiring a COBRA aggregator. For Dawid-Skene and GLAD, we additionally test controls that provide anchor labels inside the estimation procedure, and separately apply explicit anchor-based orientation to their resulting reliability orderings.

C. Anchor Protocol

For each substrate, the labelled evaluation resource is deterministically divided into a 174-item anchor pool and a disjoint evaluation set containing 1,570 SA items or 1,566 TRL pairs. Anchor items are used only for orientation and do not train or update the source models. Let m denote the number of labelled anchor items sampled from the anchor pool. The main SA grid uses $m \in \{16, 32, 64, 128, 174\}$, and the TRL grid uses $m \in \{16, 32, 64, 128\}$. Each value of m is evaluated with five independently seeded anchor draws, yielding 25 configurations per SA composition-seed cell and 20 per TRL cell. Table I averages over these complete grids. We additionally evaluate $m \in \{2, 4, 8, 16\}$ in the small-anchor ablation.

D. Anchor and Threshold Sensitivity

Let p denote the probability that an anchor label is independently inverted. We evaluate $p \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$. The grid covers eight source pools, four anchor sizes $m \in \{16, 32, 64, 128\}$, five anchor draws, and six noise rates, for $8 \times 4 \times 5 \times 6 = 960$ configurations. Corruption is uniform and random. To test the limits of this random-noise result, we additionally evaluate targeted anchor-label changes and worst-case anchor selection. These experiments test whether an informed choice of corrupted anchors can change ACT's orientation decision. We also test sensitivity to the two decision thresholds using a 10×10 grid with $\tau_s, \tau_a \in$

$\{0, 0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40\}$. The default remains $\tau_s = \tau_a = 0.05$; the sweep is used only to measure sensitivity rather than to retune the method.

E. Source-Count, Parity, and Source-Structure Sensitivity

We test sensitivity to the number of available sources by subsampling the existing nine-source pools without retraining any models. Let N' denote the number of sources in a subsampled pool and k the number of corrupted sources. We evaluate $N' \in \{3, 5, 7, 9\}$ with the following compositions:

$$\begin{aligned} N' = 3 : k \in \{1, 2\}, \quad N' = 5 : k \in \{2, 3\}, \\ N' = 7 : k \in \{2, 5\}, \quad N' = 9 : k \in \{3, 6\}. \end{aligned}$$

These settings include one trusted-majority and one corrupted-majority composition for each N' . For each composition, we evaluate up to 50 distinct source subsets, enumerating all valid subsets when fewer than 50 exist. Results are aggregated over both source-construction seeds and five anchor draws per subset. The ablation evaluates $m \in \{8, 16\}$; the reported source-count results use $m = 16$.

We extend this analysis to the omitted region around parity, evaluating corrupted-source fractions

$$k/N' \in \{0.400, 0.429, 0.500, 0.571, 0.600\}$$

at ($m=16$). This tests ACT below, at, and immediately above the trusted/corrupted majority boundary.

We also relax the two-cohort construction by evaluating source populations with two to five quality levels and a correlated-corruption setting in which corrupted sources make highly coherent errors. These experiments test whether the orientation behavior persists when source quality is graded or corruption is structured rather than produced only by the original two-cohort construction.

F. Evaluation Metrics

Trust orientation: Let $\mathbf{r} = (r_1, \dots, r_N)$ be the reliability vector produced by the evaluated estimator, and let $\mathbf{r}^{\text{decl}}$ be the known binary trusted/corrupted reference map. We report $\rho = \text{Spearman}(\mathbf{r}, \mathbf{r}^{\text{decl}})$. Positive values indicate that trusted sources receive greater reliability, whereas negative values indicate inverted orientation. Because the declared map is binary, ρ measures trust orientation.

Flip rate: A flip occurs when ACT replaces $\mathbf{a}$ with $1 - \mathbf{a}$. We report the proportion of anchor configurations with a flip; Table I reports the count. A flip records intervention, not correctness, which is assessed against Declared trust.

Downstream gain: For estimator E and aggregator G , downstream gain is $\Delta(E, G) = \text{Acc}(E, G) - \text{Acc}(\text{Equal}, \text{Averaging})$. Thus, Δ is the absolute accuracy difference from the common simple-averaging baseline.

Material-gain criterion: An ACT-aggregator configuration meets the material-gain criterion when $\Delta \geq 0.02$ and its paired-bootstrap 95% confidence interval excludes zero. Intervals use 1,000 paired resamples over evaluation items. The effect threshold was fixed before the reported evaluation grid was run. The criterion is evaluated separately for RoT, DRWA,

and AVGA, yielding $25 \times 3 = 75$ evaluations per SA cell and $20 \times 3 = 60$ per TRL cell.

VI. RESULTS

The results address three questions: whether pairwise agreement becomes inverted when corrupted sources form the majority, whether ACT corrects this inversion without reversing trusted-majority, and whether the corrected orientation improves downstream aggregation. We further examine whether this failure also appears in other reliability estimators, how sensitive ACT is to its thresholds, and how it behaves near parity and under more difficult source and anchor conditions.

A. Main Results

Table I reports results by substrate, corrupted-source count, and source-construction seed. Agreement and ACT ρ measure trust orientation, while Flip reports the number of anchor configurations in which ACT reverses Agreement. For the reported 3/6 source split, complete separation of the declared groups by a tie-free ranking gives $|\rho| \approx 0.82$, with the sign indicating whether the orientation is correct or inverted.

The downstream block reports ACT-based absolute accuracy gains Δ_G for RoT, DRWA, and AVGA and their mean,

$$\bar{\Delta}_{\text{ACT}} = \frac{\Delta_{\text{RoT}} + \Delta_{\text{DRWA}} + \Delta_{\text{AVGA}}}{3}.$$

The columns $\bar{\Delta}_{\text{Agr}}$ and $\bar{\Delta}_{\text{Equal}}$ report the corresponding means over the same aggregators. All gains are relative to Equal trust with simple averaging; 0.02 corresponds to 2 percentage points. The *Criterion met* column counts ACT-aggregator configurations that satisfy the material-gain criterion of $\Delta \geq 0.02$, with a paired-bootstrap (95%) confidence interval excluding zero.

B. Trusted-Majority Results

When 3/9 sources are corrupted, Agreement is already correctly oriented, with Agreement and ACT both achieving $\rho \approx +0.82$ in every SA and TRL cell. ACT introduces no false reversals, with 0/50 across the two SA seeds and (0/40) across the two TRL seeds. Mean downstream gains are approximately (0.01), with no configuration meeting the material-gain criterion, as indicated by (0/75) for each SA seed and (0/60) for each TRL seed. This is expected because Agreement is already correctly oriented in this regime. ACT therefore preserves the existing orientation without materially changing downstream performance.

C. Corrupted-Majority Results

When 6/9 sources are corrupted, Agreement is inverted in every main-table cell, with $\rho \approx -0.82$ on both substrates and seeds. The larger, internally coherent corrupted cohort receives higher mean pairwise agreement, so Agreement recovers the partition structure but assigns greater reliability to its corrupted side. ACT corrects this inversion in every main-table configuration, producing $\rho \approx +0.82$. It reverses Agreement in 25/25 SA anchor configurations per seed and 20/20 TRL

configurations per seed. Correcting the orientation produces substantial downstream gains. ACT gains range from +0.772 to +0.808 on SA, from +0.319 to +0.351 on TRL seed 0, and from +0.233 to +0.240 on TRL seed 1. Every corrupted-majority ACT-aggregator configuration meets the material-gain criterion, with (75/75) for each SA seed and (60/60) for each TRL seed.

The larger SA gains reflect the substantially weaker Equal-plus-averaging baseline in that controlled setting. Despite this difference in magnitude, all three reliability-aware aggregators benefit from the corrected orientation on both substrates.

D. Comparison with Declared Trust

In corrupted-majority settings, Agreement produces negative mean gains, while Equal trust is near zero or negative. ACT corrects the orientation and produces positive gains across RoT, DRWA, and AVGA. Table II shows that, at $m = 16$, ACT matches Declared trust in mean downstream gain in the evaluated binary-partition setting. This equality does not imply continuous reliability calibration or accurate within-cohort ranking.

E. Independent Reliability Estimators

Table III tests whether the corrupted-majority inversion is limited to pairwise Agreement or to the COBRA aggregation setting. Dawid-Skene, GLAD, spectral estimation, truth discovery, and a Snorkel-family label model (triplet closed form) all show negative trust orientation in the four main corrupted-majority cells. Their trusted-majority sanity checks remain positively oriented.

Applying the same anchor-based orientation step restores positive trust orientation for all five methods. At the default ($m=16$), their corrupted-majority orientation changes from approximately (-0.82) to (-0.83) before orientation to approximately (+0.82) to (+0.83) afterward. The correction is already successful from ($m=8$) in these settings.

For Dawid-Skene and GLAD, we also provide the anchor labels directly during estimation. Both methods remain inverted, even with this supervision. This shows that simply providing labeled examples does not resolve the orientation ambiguity; the explicit orientation step is still needed.

F. Anchor Budget

We evaluate $m \in \{2, 4, 8, 16\}$ on both substrates. In corrupted-majority SA settings, ACT correctly reorients at every anchor size. For TRL, reorientation is seed-dependent at $m \in \{2, 4\}$, but succeeds for every evaluated anchor draw at $m \in \{8, 16\}$. We therefore use $m = 16$ as a conservative shared setting for the source-count analysis.

The material-gain criterion is met by every corrupted-majority SA configuration and every corrupted-majority TRL configuration at $m \in \{8, 16\}$. For TRL, however, the pass rate falls to 70% at $m = 2$ and 60% at $m = 4$, consistent with ACT's orientation instability at the two smallest anchor sizes.

At $m = 16$, ACT matches the Probe on SA (0.784/0.799) and gives higher mean downstream gains on

TABLE I
NINE-SOURCE RESULTS. RoT, DRWA, AND AVGA USE ACT RELIABILITIES; MEAN COLUMNS AVERAGE THE SAME AGGREGATORS. GAINS ARE RELATIVE TO EQUAL+AVERAGING. NEGATIVE ρ DENOTES INVERSION; FLIP COUNTS ACT REVERSALS.

Substrate	Corrupted	Seed	Trust orientation			Gain using ACT reliabilities				Mean gain by trust map		
			Agreement ρ	ACT ρ	Flip	RoT	DRWA	AVGA	$\bar{\Delta}_{ACT}$	$\bar{\Delta}_{Agr}$	$\bar{\Delta}_{Equal}$	Criterion met
SA	3/9	0	+0.82	+0.82	0/25	+0.011	+0.010	+0.010	+0.010	+0.010	+0.000	0/75
		1	+0.82	+0.82	0/25	+0.008	+0.010	+0.013	+0.010	+0.010	+0.000	0/75
	6/9	0	-0.82	+0.82	25/25	+0.772	+0.790	+0.789	+0.784	-0.020	+0.001	75/75
		1	-0.82	+0.82	25/25	+0.785	+0.805	+0.808	+0.799	-0.010	+0.000	75/75
TRL	3/9	0	+0.82	+0.82	0/20	+0.006	+0.006	+0.015	+0.009	+0.009	+0.002	0/60
		1	+0.82	+0.82	0/20	+0.014	+0.013	+0.008	+0.012	+0.012	+0.002	0/60
	6/9	0	-0.82	+0.82	20/20	+0.319	+0.338	+0.351	+0.336	-0.050	-0.012	60/60
		1	-0.82	+0.82	20/20	+0.233	+0.239	+0.240	+0.237	-0.150	-0.047	60/60

TABLE II
MEAN CORRUPTED-MAJORITY GAIN $\bar{\Delta}$ OVER RoT, DRWA, AVGA, AND BOTH SEEDS; ACT USES $m = 16$.

Method	Trust supervision	SA $\bar{\Delta}$	TRL $\bar{\Delta}$
Agreement	None	-0.015	-0.100
ACT	16 labelled anchors	+0.791	+0.287
Declared	Full trust partition	+0.791	+0.287

TABLE III
INDEPENDENT RELIABILITY ESTIMATORS UNDER CORRUPTED-MAJORITY FEEDBACK. NEGATIVE ρ INDICATES INVERTED TRUST ORIENTATION.

Method	Before orientation ρ	After orientation ρ
Dawid-Skene	$[-0.83, -0.82]$	$[+0.82, +0.83]$
GLAD	$[-0.83, -0.82]$	$[+0.82, +0.83]$
Spectral	$[-0.83, -0.82]$	$[+0.82, +0.83]$
Truth discovery	-0.82	+0.82
Snorkel-family triplet	$[-0.83, -0.82]$	$[+0.82, +0.83]$

TRL (0.336/0.237 versus 0.256/0.228). At $m \in \{2, 4\}$, six of 20 TRL Probe configurations yield undefined Spearman correlation due to constant reliability maps. Neither method uniformly dominates: Probe can give stronger rank correlation, while ACT generally gives higher TRL downstream gains.

G. Threshold Sensitivity

We next test whether the default $\tau_s = \tau_a = 0.05$ is sensitive to a particular threshold choice. Across the 10×10 sweep, all main corrupted-majority cells retain a positive ACT orientation for every pair with $\tau_s, \tau_a \leq 0.20$. At $m = 16$, the smallest agreement-separation margin is 0.241, and the smallest absolute anchor margin is 0.115, both of which exceed the default threshold of 0.05. Larger thresholds eventually stop ACT from making reversals that are otherwise supported by the anchors, with this effect appearing most clearly on TRL. The results therefore show that 0.05 lies inside a broad stable region rather than at a narrow tuned operating point.

H. Random Anchor-Label Noise

We test sensitivity to random anchor-label corruption at $p \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$, where p is the probabil-

TABLE IV
RANDOM ANCHOR-LABEL NOISE SENSITIVITY. ACT RETAINS AT LEAST 97.5% CORRECT ORIENTATION THROUGH $p = 0.3$, DEGRADES AT $p = 0.4$, AND BECOMES UNINFORMATIVE AT $p = 0.5$.

Noise p	Correct orientation	False flip	Mean Δ
0.0	1.000	0.000	+0.547
0.1	1.000	0.000	+0.547
0.2	0.975	0.000	+0.538
0.3	0.988	0.000	+0.543
0.4	0.787	0.000	+0.444
0.5	0.263	0.263	+0.103

ity that each binary anchor label is independently inverted. Table IV summarizes 80 corrupted-majority and 80 trusted-majority configurations at each noise rate. ACT remains stable under moderate random anchor-label noise, retaining correct corrupted-majority orientation in at least 97.5% of configurations through $p = 0.3$, with no false trusted-majority reversals. Performance drops at $p = 0.4$ and becomes uninformative near $p = 0.5$, as shown in Table IV. These results demonstrate tolerance to independent random label noise, but not to adversarial anchor corruption.

We test this limitation directly using targeted label changes and worst-case anchor selection. At ($m=16$), worst-case anchor selection reduces correct corrupted-majority orientation to 0% on both substrates. Under targeted label flips, TRL falls to 40% correct orientation at 10% corruption and to 0% at 30%. SA is more resistant, remaining correctly oriented through 30% corruption, but falls to 30% at 40% corruption and to 0% at 50%. Thus, tolerance to random anchor noise should not be interpreted as robustness to an informed adversary.

I. Source-Count and Near-Parity Sensitivity

Figure 2 reports ACT and Agreement orientation after post-hoc subsampling of the nine-source pools. Across corrupted-majority cells, ACT flips Agreement at a rate of at least 0.953, reaching 1.000 except for TRL at $N' = 3$. Across trusted-majority cells, the flip rate is at most 0.002, so ACT preserves the correct orientation. Mean downstream gain remains positive in every corrupted-majority cell, ranging from about

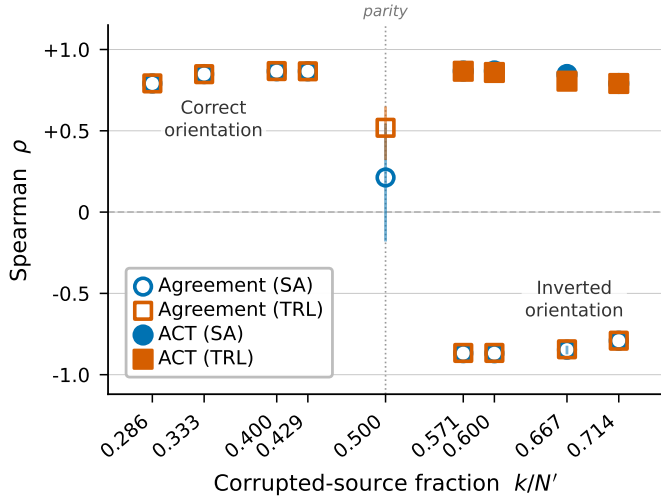


Fig. 2. Agreement ρ (open) and ACT ρ (filled) versus corrupted-source fraction k/N' at $m = 16$. Agreement remains correctly oriented below parity and becomes inverted above parity; ACT preserves or restores the correct orientation. At exact parity, the separability check prevents a reversal.

+0.40 to +0.80 for SA and +0.18 to +0.34 for TRL. Because this study subsamples existing pools without retraining, it tests robustness to source removal rather than generalization.

To cover the near-parity region omitted from the original sweep, we separately evaluate $k/N' \in \{0.400, 0.429, 0.500, 0.571, 0.600\}$ at $m = 16$. Below parity, Agreement remains correctly oriented and ACT preserves it. Above parity, Agreement becomes inverted, while ACT restores the correct orientation in all cells at 0.571, and in 100% of SA and 99.5% of TRL evaluations at 0.600. At exact parity (0.500), all cases fail the separability check, so ACT makes no reversal rather than forcing an unsupported trust direction.

J. More Complex Source Conditions

ACT still works when there is a clear group of badly corrupted sources, because it can use the anchors to decide which side should be trusted. But when source quality is spread gradually across good, medium, and bad sources, ACT 0.47 is not very good at ranking each source by exact reliability, which is why Dawid-Skene 0.94 performs better there. When corrupted sources make the same mistakes together, Agreement can strongly trust the wrong group, while ACT can flip the trust direction back correctly. Overall, ACT is mainly useful for deciding which side to trust, not for giving every source a precise reliability score.

VII. CONCLUSION

Pairwise agreement misassigns trust to the wrong cohort when corrupted sources dominate. ACT resolves this ambiguity by orienting agreement-derived cohorts using a small labeled anchor set. Across SST-2 classification and HH-RLHF preference modeling, ACT maintains trusted majority orientation and corrects corrupted-majority inversion, yielding significant downstream gains. This orientation failure affects

five other reliability-estimation methods, while anchor-based orientation restores correct direction. ACT remains stable across diverse thresholds and behaves as intended near the majority boundary, abstaining from reversal at exact parity when cohorts lack sufficient separation. Graded-source, correlated-corruption, and adversarial scenarios further define its scope: ACT distinguishes as a trust-orientation method, not a general reliability estimator or adversarially robust anchor mechanism. Future work will address secure anchor mechanisms and natural source corruption.

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IX. AI USE ACKNOWLEDGMENT

Generative AI and language-editing tools (e.g. ChatGPT, Claude, Grammarly) were used for grammar correction and language refinement. All edited content was reviewed and verified by the authors, who take full responsibility for the final manuscript.

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